



## Modelling Financial Markets

Chapter 20 (special chapter): Path Signatures and Realised Volatility — a  
Research Case Study



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## Acronyms used in this material (1/2)

- ▣ AAPL: Apple Inc. (Nasdaq ticker)
- ▣ ACM: Association for Computing Machinery
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ AUD: Australian Dollar (ISO 4217 code)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BIC: Bayesian Information Criterion
- ▣ BPQ: Bollerslev–Patton–Quaedvlieg (insanity filter of the HARQ paper)
- ▣ CAD: Canadian Dollar (ISO 4217 code)
- ▣ CL: Crude oil (WTI light sweet) futures, NYMEX ticker
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: E-mini S&P 500 futures (CME ticker)
- ▣ EURUSD: Euro / US dollar exchange rate (FX pair code)
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ FX: Foreign eXchange
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GC: Gold futures, COMEX ticker
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HARQ: HAR model with Realised Quarticity
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)
- ▣ KDD: Knowledge Discovery and Data Mining (ACM SIGKDD conference)

## Acronyms used in this material (2/2)

- LARS: Least-Angle RegreSsion (algorithm)
- LASSO: Least Absolute Shrinkage and Selection Operator
- MCS: Model Confidence Set
- MSE: Mean Squared Error
- NG: Natural Gas futures, NYMEX ticker
- NVDA: NVIDIA Corporation (Nasdaq ticker)
- OLS: Ordinary Least Squares
- PDE: Partial Differential Equation
- PRDS: Positive Regression Dependence on a Subset (of p-values)
- QLIKE: Quasi-LIKelihood (loss function)
- RQ: Realised Quarticity
- RV: Realised Variance
- SHAR: Semivariance HAR model
- TSLA: Tesla, Inc. (Nasdaq ticker)
- US: United States
- USDJPY: US dollar / Japanese yen exchange rate (FX pair code)
- VOLARE: VOLatility Archive for Realized Estimates (database)
- XOM: Exxon Mobil Corporation (NYSE ticker)

## Chapter Route and Outcomes

### Route of this chapter

- A research case study: the method of [Gu et al. \(2024\)](#), published for freight rates at Amazon, transferred to daily realised variance of 50 VOLARE assets [Cipollini et al. \(2026\)](#)
- Part I: realised measures, the data, the HAR benchmarks, path signatures
- Part II: the paper of Gu et al., a design fixed before the results, results, robustness, one pass of the AI-assisted discovery loop
- Implementation detail sits in the [Appendix](#) at the end

### After this chapter you can

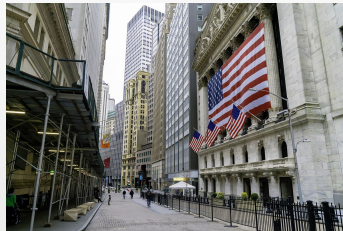
- compute the signature of a piecewise-linear path and read its Lévy area
- build signature features and kernel weights for a volatility forecast
- compare forecasts on many assets with DM tests, model confidence sets and Holm/BH corrections
- pre-register a method transfer and report a null result with its mechanism

## Why Forecast Realised Variance?

- Volatility is an input to almost every risk measure in this course
  - ▶ Tail-risk measures ([Chapter 7](#)), option prices ([Chapter 12](#)), portfolio weights ([Chapter 4](#))
- Intraday data make daily variance almost observable: realised variance (RV, [Chapter 9](#))
  - ▶ So volatility forecasting becomes a regression problem with a noisy but model-free target
- The benchmark is hard to beat: the HAR model [Corsi \(2009\)](#) ([Chapter 9](#))
  - ▶ Three averages of past RV; linear; estimated by OLS in a second
- Any new method must beat HAR out of sample, on many assets, with a proper test

## The Research Question

- Gu et al. describe a recent window of a series as a **path** and summarise it by its **signature**
  - ▶ Signature features enter a sparse linear regression (two-step LASSO)
  - ▶ Past periods whose path looks like today's get more weight (a signature kernel)
- They report a more than fivefold accuracy gain over an industry forecast of freight rates
  - ▶ Question: does the same recipe improve on HAR-type models for realised variance?
- Two separate ingredients to test: the signature features and the kernel weights



Wall Street and the New York Stock Exchange, 25  
March 2020

Photo: [Billie Grace Ward \(2020\)](#); CC BY 2.0; [Wikimedia Commons](#)

## Realised Variance and Quadratic Variation

- Log price  $p_t$  with  $dp_t = \mu_t dt + \sigma_t dW_t + dJ_t$ ; day  $t$  split into  $M$  intraday returns  $r_{t,i}$ 
  - ▶  $RV_t = \sum_{i=1}^M r_{t,i}^2 \xrightarrow{M \rightarrow \infty} \int_{t-1}^t \sigma_s^2 ds + \sum_{t-1 < s \leq t} (\Delta J_s)^2$  [Barndorff-Nielsen & Shephard \(2002\)](#)
- Without jumps and noise, on an equally spaced grid, the limit is the integrated variance  $IV_t$ ; the error is mixed Normal
  - ▶  $\sqrt{M}(RV_t - IV_t) \Rightarrow MN(0, 2IQ_t)$ ,  $IQ_t = \int_{t-1}^t \sigma_s^4 ds$  (integrated quarticity)
- Microstructure noise makes very fine sampling biased; five-minute returns are the standard compromise ([Chapter 9](#))
- [Chapter 9](#) derives these results; they are used here as given

## The Realised Measures in VOLARE

- Semivariances [Barndorff-Nielsen, Kinnebrock & Shephard \(2010\)](#):  $RV_t^+ = \sum_i r_{t,i}^2 1\{r_{t,i} > 0\}$ ,  
 $RV_t^- = \sum_i r_{t,i}^2 1\{r_{t,i} < 0\}$ 
  - ▶ They add up:  $RV_t = RV_t^+ + RV_t^-$  (holds in VOLARE to machine precision)
- Realised quarticity:  $RQ_t = \frac{M}{3} \sum_i r_{t,i}^4 \rightarrow IQ_t$ , the scale of the measurement error of RV
  - ▶ The limit needs a continuous, noise-free price and an equally spaced grid; a jump  $\Delta J$  adds about  $M(\Delta J)^4/3$ , so raw RQ is jump-sensitive
- Bipower variation (robust to jumps), realised kernel (robust to noise) [Barndorff-Nielsen et al. \(2008\)](#), range-based estimators
- Target of this chapter: five-minute RV
  - ▶ Across 400 assets, little beats five-minute RV as a proxy of daily variance [Liu, Patton & Sheppard \(2015\)](#)
  - ▶ Its semivariances and quarticity are on the same grid, so SHAR and HARQ use consistent inputs

## Targets and Losses

- Horizon  $h$ :  $RV_{t+1:t+h} = \frac{1}{h} \sum_{j=1}^h RV_{t+j}$ ,  $h \in \{1, 5, 22\}$ ; log models forecast  $\ln RV_{t+1:t+h}$ 
  - ▶ Back to levels with  $F_t = \exp(\hat{y}_t + \hat{\sigma}^2/2)$ ,  $\hat{\sigma}^2$  = residual variance: exact if the log error is conditionally Normal with constant variance, otherwise an approximation
  - ▶ For  $h > 1$  targets overlap: this induces serial dependence (through lag  $h - 1$  by construction, possibly longer), so every test uses a HAC variance
- A loss is **robust** if, for a conditionally unbiased proxy ( $\mathbb{E}_{t-1}[RV_t] = \sigma_t^2$ ), the *expected*-loss ranking with  $RV_t$  equals that with the true variance [Patton \(2011\)](#)
  - ▶  $MSE_t = (RV_t - F_t)^2$ ,  $QLIKE_t = RV_t/F_t - \ln(RV_t/F_t) - 1 \geq 0$  (zero only if  $F_t = RV_t$ )
- QLIKE depends on the ratio  $RV_t/F_t$ : the same proportional error has the same loss on a calm day as on a stormy day
- MSE is dominated by a few crisis days; QLIKE is the primary loss, MSE the secondary one

## VOLARE: an Open Archive of Realised Measures

- Built at the Universities of Messina and Florence from tick data [Cipollini et al. \(2026\)](#)
  - ▶ Asset-specific cleaning, trading calendars and timestamp precision; the same estimators for every asset class
- Free for research at [volare.unime.it](http://volare.unime.it); not redistributed with this course
  - ▶ Download the realised-variance files and point the notebooks to your local folder
- Citation requested by the authors: Cipollini, Cruciani, Gallo, Insana, Otranto & Spagnolo (2026), arXiv:2602.19732



University of Messina, the home of VOLARE

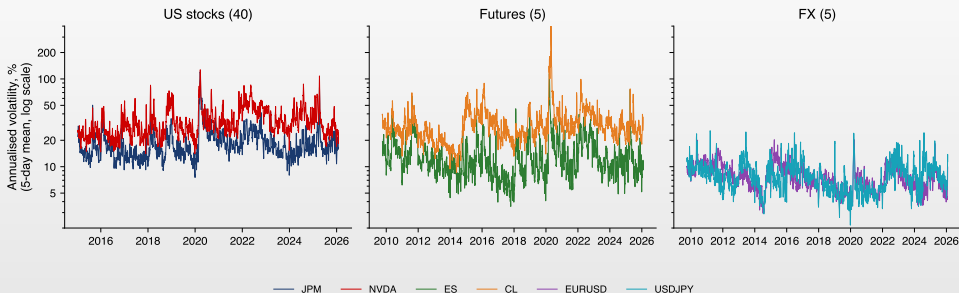
Photo: [Superchilum \(2013\)](#); CC BY-SA 3.0; [Wikimedia Commons](#)

## The Panel: 50 Assets in Three Classes

| Class                               | Assets | Sample                    | Observations | Median volatility |
|-------------------------------------|--------|---------------------------|--------------|-------------------|
| American large-cap stocks           | 40     | 2 Jan 2015 – 30 Jan 2026  | 111 440      | 17.1%             |
| Futures (C, CL, ES, GC, NG)         | 5      | 28 Sep 2009 – 30 Jan 2026 | 21 059       | 19.7%             |
| FX (AUD, EUR, GBP, CAD, JPY vs USD) | 5      | 25 Sep 2009 – 30 Jan 2026 | 21 203       | 7.3%              |

- Stocks: 40 members of the Dow Jones 30 and S&P 100 (AAPL to XOM); futures: corn, crude oil, E-mini S&P 500, gold, natural gas
- Median volatility:  $\sqrt{252 \text{ median}(\text{RV}_t)}$  per asset, then the median over the class; the data contain trading days only
- Variables used: five-minute RV, its positive and negative semivariances ( $\text{RV}^+$ ,  $\text{RV}^-$ ), realised quarticity (RQ), and the daily open and close prices

## Realised Volatility Across Asset Classes



- Annualised volatility  $\sqrt{252 RV_t}$ , five-day mean, log scale, two assets per class (stocks: JPM = JPMorgan Chase, NVDA = Nvidia; futures: ES = E-mini S&P 500, CL = crude oil; FX: EURUSD, USDJPY = US dollar in yen); typical levels: FX about 7.3%, stocks about 17.1% a year
- All classes share March 2020; crude oil (CL) peaks in April 2020, when the front contract traded below zero: a forecaster must adapt within days
- On a log scale volatility is stationary and persistent (the long memory HAR approximates); spikes decay within weeks, so the shape of the recent path carries information

## HAR: Three Averages of the Past

- HAR (Heterogeneous AutoRegressive) model [Corsi \(2009\)](#)

- ▶  $RV_{t+1:t+h} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$
- ▶  $RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}$ ,  $RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$

- HAR is a restricted AR(22): the 22 lag coefficients take only three distinct values

- ▶  $\phi_1 = \beta_d + \beta_w/5 + \beta_m/22$ ,  $\phi_{2..5} = \beta_w/5 + \beta_m/22$ ,  $\phi_{6..22} = \beta_m/22$

- Traders at daily, weekly and monthly frequencies: a cascade that mimics long memory with three parameters

- log-HAR:  $\ln RV_{t+1:t+h}$  on  $\ln RV_t$ ,  $\ln RV_t^{(w)}$ ,  $\ln RV_t^{(m)}$ ; no negative forecasts, less driven by spikes

- ▶ Logs of averages are not averages of logs, so this log-HAR is not an exact restricted AR(22) in  $\ln RV$  (Seminar 20, A5)

## Two Refinements: HARQ and SHAR

- RV is a generated regressor:  $RV_t = IV_t + \eta_t$ ,  $\text{Var}(\eta_t) \approx 2IQ_t/M$ ; errors-in-variables attenuate  $\beta_d$  on noisy days
  - ▶ HARQ [Bollerslev, Patton & Quaedvlieg \(2016\)](#):  
 $RV_{t+1:t+h} = \beta_0 + (\beta_d + \beta_Q(RQ_t^{1/2} - \overline{RQ^{1/2}}))RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$ , expected  $\beta_Q < 0$
  - ▶ BPQ filter: a forecast outside the in-sample range of the target is replaced by the in-sample mean (“insanity filter”)
- SHAR [Patton & Sheppard \(2015\)](#): split the daily term by the sign of the intraday returns
  - ▶  $RV_{t+1:t+h} = \beta_0 + \beta^+ RV_t^+ + \beta^- RV_t^- + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$
  - ▶ Patton and Sheppard find  $\beta^- > \beta^+$ : an intraday version of the leverage effect ([Chapter 5](#))
- The signature can capture a related effect from daily data: the Lévy area between returns and log RV (Path Signatures section)

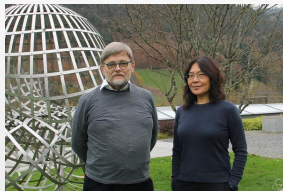
## Seven Models in One Table

| Model     | Target   | Weights          | Regressors                                      |
|-----------|----------|------------------|-------------------------------------------------|
| HAR       | RV       | equal            | $RV_t, RV_t^{(w)}, RV_t^{(m)}$                  |
| log-HAR   | $\ln RV$ | equal            | the same, in logs                               |
| HARQ      | RV       | equal            | $HAR + RV_t (RQ_t^{1/2} - \overline{RQ^{1/2}})$ |
| SHAR      | RV       | equal            | $RV_t^+, RV_t^-, RV_t^{(w)}, RV_t^{(m)}$        |
| log-HAR-K | $\ln RV$ | signature kernel | log-HAR regressors                              |
| Sig-L     | $\ln RV$ | equal            | log-HAR + 39 signature terms, two-step LASSO    |
| Sig-LK    | $\ln RV$ | signature kernel | as Sig-L: the method of Gu et al.               |

- The last four form a  $2 \times 2$  ablation (each ingredient switched on and off separately): signature features (yes/no)  $\times$  kernel weights (yes/no)
- Log-HAR is the natural baseline for the signature models: same target, and its regressors are among the candidates; but the LASSO may also drop log-HAR terms, so Sig-L vs log-HAR compares two procedures, not only two feature sets

## Where Signatures Come From

- **Chen (1957)**: iterated integrals of a path form an algebra and determine it up to reparametrisation
  - ▶ Kuo-Tsai Chen, geometry of loop spaces
- **Lyons (1998)**: rough path theory; the signature is the natural feature set for controlled differential equations
  - ▶ A response to a path depends on it (to first order) only through its signature
- **Machine learning**: [Levin, Lyons & Ni \(2013\)](#), [Chevyrev & Kormilitzin \(2016\)](#), [Lyons & McLeod \(2022\)](#); finance: signature-based asset price models [Cuchiero, Gazzani & Svaluto-Ferro \(2023\)](#)



Terry Lyons (left), Oberwolfach, 2024

Photo: [Katrin Schmid \(2024\)](#); CC BY-SA 2.0 DE; [Wikimedia Commons](#)

## A Window of Data Is a Path

- Take the last  $L$  days of  $d - 1$  series and connect the points linearly
  - ▶  $X : [0, 1] \rightarrow \mathbb{R}^d$ ,  $X_{u_i} = (u_i, \ln \text{RV}_{t-L+1+i}, \sum_{j \leq t-L+1+i} r_j)$ ,  $u_i = i/(L - 1)$
- **Time augmentation:** the first channel is time itself
  - ▶ Without it the signature cannot tell a fast move from a slow one (reparametrisation invariance)
- A regression on the window sees 22 lags; a path summary sees their order, speed and co-movement
  - ▶ Example: RV rose while prices fell, or RV rose first and prices fell later

## The Signature: Definition

- For a path  $X : [a, b] \rightarrow \mathbb{R}^d$  of bounded variation and a word  $(i_1, \dots, i_k)$  with letters in  $\{1, \dots, d\}$ 
  - ▶  $S(X)_{a,b}^{i_1 \dots i_k} = \int_{a < u_1 < \dots < u_k < b} dX_{u_1}^{i_1} \dots dX_{u_k}^{i_k}$  (iterated integral)
- The signature is the whole sequence of levels
  - ▶  $\text{Sig}(X) = (1, S^{(1)}, S^{(2)}, S^{(3)}, \dots)$ ,  $S^{(k)} \in (\mathbb{R}^d)^{\otimes k}$  has  $d^k$  terms
- Level 1:  $S^i = X_b^i - X_a^i$ , the total increments
  - ▶ Level 2:  $S^{ij} = \int_a^b (X_u^i - X_a^i) dX_u^j$ ; level 3 adds one more integral

## Linear Segments and Chen's Identity

- A linear segment with increment  $\Delta \in \mathbb{R}^d$  has the tensor exponential as signature
  - ▶  $\text{Sig}(\text{segment}) = \exp_{\otimes}(\Delta) = (1, \Delta, \frac{1}{2!}\Delta^{\otimes 2}, \frac{1}{3!}\Delta^{\otimes 3}, \dots)$
- **Chen's identity** [Chen \(1957\)](#): the signature of a concatenation is the tensor product
  - ▶  $\text{Sig}(X * Y) = \text{Sig}(X) \otimes \text{Sig}(Y), \quad (A \otimes B)^{(k)} = \sum_{p=0}^k A^{(k-p)} \otimes B^{(p)}$
- A piecewise-linear path with  $m$  segments:  $\text{Sig} = \exp_{\otimes}(\Delta_1) \otimes \dots \otimes \exp_{\otimes}(\Delta_m)$ 
  - ▶ This is the whole algorithm:  $m$  updates, each costing  $O(d^N)$  at depth  $N$  (a few lines of numpy)
  - ▶ Rolling windows: Chen also gives  $\text{Sig}(X_{[s,t]}) = \text{Sig}(X_{[a,s]})^{-1} \otimes \text{Sig}(X_{[a,t]})$

## Worked Example: a Two-Segment Path

### Path

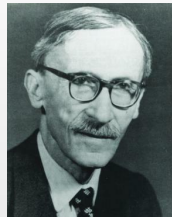
- ▣  $X : (0, 0) \rightarrow (1, 0) \rightarrow (1, 2); \Delta_1 = (1, 0), \Delta_2 = (0, 2)$
- ▣ Level 1:  $S^{(1)} = \Delta_1 + \Delta_2 = (1, 2)$
- ▣ Level 2 by Chen:  $S^{ij} = \frac{1}{2}\Delta_1^i\Delta_1^j + \Delta_1^i\Delta_2^j + \frac{1}{2}\Delta_2^i\Delta_2^j$

### Result

- ▣  $S^{11} = \frac{1}{2}, S^{12} = 0 + 1 \cdot 2 + 0 = 2$
- ▣  $S^{21} = 0, S^{22} = \frac{1}{2} \cdot 2^2 = 2$
- ▣ Check:  $S^{11} = (\Delta X^1)^2/2$  and  $S^{22} = (\Delta X^2)^2/2$  (shuffle identity)
- ▣  $S^{12} \neq S^{21}$ : the order matters (first right, then up)

## The Lévy Area: Order as a Number

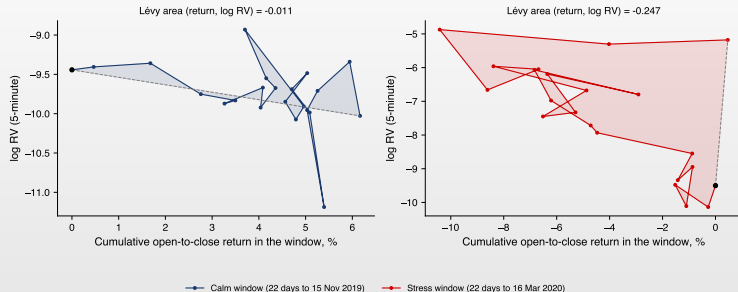
- Antisymmetric part of level 2
  - ▶  $A^{ij} = \frac{1}{2} (S^{ij} - S^{ji}) =$  signed area between the path and its chord
- Symmetric part:  $\frac{1}{2} (S^{ij} + S^{ji}) = \frac{1}{2} \Delta X^i \Delta X^j$ , no new information
  - ▶ Example above:  $A^{12} = (2 - 0)/2 = 1$ , the area of the triangle under the path
- Channels (return, log RV): a price fall followed by a volatility rise gives  $A < 0$ , but so does a rise followed by a volatility fall
  - ▶ A lead-lag statistic: leverage-type sequences ([Chapter 5](#)) are one contributor to a negative area; the sign alone does not identify them



Paul Lévy (1886–1971)

Photo: Konrad Jacobs; CC BY-SA 2.0 DE;  
[Wikimedia Commons](#)

## Two Windows of JPM as Paths

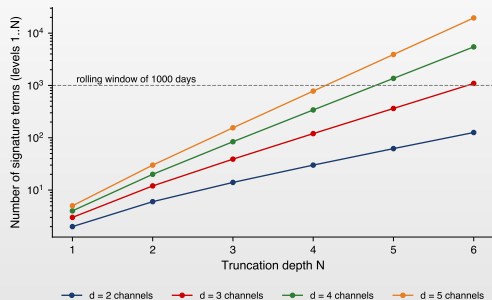


- Path (cumulative open-to-close return, log RV) over 22 days; dashed chord from start to end; the shaded region gives the Lévy area
- Calm window (to 15 Nov 2019): log RV moves by -0.58, the path wanders, Lévy area -0.011; stress window (to 16 Mar 2020): log RV rises by +4.32 ( $RV \times 75$ ) while returns fall, Lévy area -0.247
- The net intraday return is only +0.5%: the crash was largely overnight, outside this channel; no signature level recovers what the input omits

## Why Signatures Work: Three Theorems

- **Uniqueness** [Hamblly & Lyons \(2010\)](#): the full  $\text{Sig}(X)$  determines  $X$  up to tree-like equivalence and translation; with a strictly increasing time channel, up to translation only
  - ▶ A finite truncation  $\text{Sig}^N$  is not injective
- **Shuffle identity**: products of signature terms are linear combinations of higher terms,  $S^i S^j = S^{ij} + S^{ji}$ ,  
 $S^{ij} S^k = S^{ijk} + S^{ikj} + S^{kij}$ 
  - ▶ So a linear model in  $\text{Sig}^N(X)$  already contains polynomials of the increments
- **Universality**: for a compact set  $K$  (1-variation topology) of time-augmented paths with a common start and  $f : K \rightarrow \mathbb{R}$  continuous, for every  $\varepsilon > 0$  there is a linear  $\ell$  with  $\sup_{X \in K} |f(X) - \langle \ell, \text{Sig}(X) \rangle| < \varepsilon$  (Stone–Weierstrass, via the shuffle identity)
  - ▶ This is why a LASSO on signatures can approximate nonlinear functions of the path; the price is  $d^N$  features, so sparsity is needed
- **Factorial decay**:  $\|S^{(k)}\| \leq \|X\|_{1\text{-var}}^k / k!$ : truncating at a small depth  $N$  loses little for paths of moderate length

## How Many Features? The Truncation Dimension



- Levels 1.. $N$  of a  $d$ -dimensional path:  

$$M(d, N) = \sum_{k=1}^N d^k = \frac{d^{N+1} - d}{d - 1} \text{ terms}$$
- This chapter:  $d = 3$  channels (time, log RV, cumulative return), depth  $N = 3$ :  $3 + 9 + 27 = 39$  terms
  - ▶ Three are constant (words with only the time letter), so 36 vary
- $d = 5$ ,  $N = 4$  would give 780 terms for 1000 observations: the LASSO would select from noise
- Gu et al. do not disclose  $N$  and  $L$ ; here  $N = 3$ ,  $L = 22$  are fixed ex ante, with  $N = 2$  as a robustness check

## Two Invariances to Keep in Mind

- ▣ **Translation invariance:**  $\text{Sig}(X + c) = \text{Sig}(X)$  — only increments enter
  - ▶ The signature does not see the **level** of volatility, only the shape of its recent path
  - ▶ So the level must enter separately: the log-HAR terms play the role of the external factors  $x_T$  of Gu et al.
- ▣ **Reparametrisation invariance:** running the path faster or slower leaves Sig unchanged
  - ▶ Removed by the time channel; basepoint augmentation (prepending a segment from 0 to the original starting point) would remove the first invariance [Morrill et al. \(2020\)](#)
  - ▶ Translating the path to start at 0 does not: it is itself a translation (Seminar 20, C1 uses the prepended segment)

## The Signature Kernel

- Kernel = inner product of feature maps [Király & Oberhauser \(2019\)](#)
  - ▶  $k_N(X, Y) = \langle \text{Sig}^N(X), \text{Sig}^N(Y) \rangle = 1 + \sum_{k=1}^N \sum_{|w|=k} S(X)^w S(Y)^w$
- Distance induced by the kernel (Gu et al., Eq. 9)
  - ▶  $d_{\text{Sig}}(X, Y) = k(X, X) - 2k(X, Y) + k(Y, Y) = \|\text{Sig}^N(X) - \text{Sig}^N(Y)\|^2$
- Gu et al. and this chapter use this **truncated** kernel, with  $N$  as for the features
  - ▶ The untruncated kernel solves a Goursat PDE (partial differential equation) [Salvi et al. \(2021\)](#); the paper cites it but does not use it
- Channels are rescaled before the kernel ( [Appendix](#) ): otherwise the channel with the largest units dominates

## The Problem at Amazon and the Features

- Weekly North American truckload rates, national and five regions, 2018–2022, forecast up to 12 weeks ahead [Gu et al. \(2024\)](#)
  - ▶ More than 100 candidate factors screened to about 40; in Amazon operations since November 2022; KDD 2024 (ACM Conference on Knowledge Discovery and Data Mining)
- Features at week  $\tau$  (Section 4.1, Eq. 14): external factors  $x_\tau$  and the signature of the time-augmented target path
  - ▶  $\xi_\tau = [x_\tau, \text{Sig}^N(\hat{y}_{\tau-l:\tau})]$ ,  $\hat{y}_{\tau-l:\tau} = \{(t_i, y_{t_i})\}$
- One direct regression per horizon,  $y_{\tau+\Delta t}$  on  $\xi_\tau$ ; lookback  $l$ , depth  $N$ , temperature  $\gamma$  are not published (confidentiality)



A container truck on an American interstate, 2025

Photo: [DanTD \(2025\)](#); CC BY 4.0; [Wikimedia Commons](#)

## The Method: Two-Step LASSO

- ▣ Selection by a weighted LASSO (Least Absolute Shrinkage and Selection Operator) [Tibshirani \(1996\)](#) (Eq. 14)
  - ▶  $\hat{\theta}_L = \arg \min_{\theta} \sum_{\tau} w_{\tau} (y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta)^2 + \lambda \|\theta\|_1$
- ▣ Refit by weighted OLS on the selected support (Eq. 15)
  - ▶  $\hat{\theta} = \arg \min_{\text{supp}(\theta)=\text{supp}(\hat{\theta}_L)} \sum_{\tau} w_{\tau} (y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta)^2$
- ▣ Why refit: LASSO shrinks the kept coefficients towards zero; OLS after selection removes that bias [Belloni & Chernozhukov \(2013\)](#)
  - ▶ The paper calls it “adaptive” because of the weights  $w_{\tau}$ , not in the sense of the adaptive LASSO of [Zou \(2006\)](#)
- ▣ How  $\lambda$  is chosen is not reported

## The Method: Kernel Weights

- ▣ Kernel path: the *pairs*  $z_\tau = (x_\tau, y_{\tau+\Delta t})$  of predictors and later target, not the features  $\xi_\tau$  (Algorithm 1)
  - ▶ Distance to the last window with an observed target (Eq. 11):  $\delta_\tau = d_{\text{Sig}, N}(z_{\tau-l:\tau}, z_{t-\Delta t-l:t-\Delta t})$
- ▣ Softmax weights with temperature  $\gamma \geq 0$  (Eq. 12)
  - ▶  $w_\tau = \frac{\exp(-\gamma \delta_\tau)}{\sum_s \exp(-\gamma \delta_s)}$
  - ▶  $\gamma = 0$ : ordinary (equal-weight) regression;  $\gamma \rightarrow \infty$ : only the most similar past window
- ▣ A nearest-neighbour idea inside a regression: learn from the past periods that looked like today
  - ▶ Cost of the idea: fewer effective observations,  $n_{\text{eff}} = 1 / \sum_\tau w_\tau^2$  (Kish)

## What the Paper Reports, Read Critically

- National rates, three months ahead, April–November 2021 (Table 1): relative error  $|y - \hat{y}|/|y|$  of about 1–2% vs 13–27% for the industry forecast, “more than fivefold”
  - ▶ Horizons 1–12 weeks (Table 2): 1.06–2.60% nationally, 1.05–5.69% by region; estimated annual saving of about 50 million dollars
  - ▶ Kernel ablation (Table 3, six weekly values, 31 October–5 December 2021, one region): errors up to 9.85% without the kernel vs up to 5.15% with it
- Limits of the evidence
  - ▶ **One benchmark:** an unnamed commercial forecast; no ARIMA, no LASSO without signatures, no random walk
  - ▶ **Short evaluation:** a few months, a few series, no standard errors, no test of equal accuracy
  - ▶ **Undisclosed tuning:**  $l$ ,  $N$ ,  $\gamma$ ,  $\lambda$  and how they were chosen
- None of this makes the method wrong; it makes the evidence thin, which is why it is tested elsewhere, replication before extension ([Chapter 19](#))

## What Transfers to Volatility, and What Does Not

### Transfers

- ▣ Signature of the time-augmented recent path of the target
- ▣ Two-step LASSO with weights, direct regression per horizon
- ▣ Softmax weights  $\exp(-\gamma \delta_\tau)$  with the truncated signature kernel

### Must be adapted

- ▣ Daily, noisy target instead of weekly rates: logs, HAC inference, robust losses
- ▣ External factors  $x_\tau$  replaced by the log-HAR terms (they carry the level)
- ▣ Undisclosed choices  $(l, N, \gamma, \lambda)$  fixed ex ante in this chapter
- ▣ Kernel: the path of this chapter (time,  $\ln RV$ , return) ending at  $\tau$  vs the window ending at  $t$ , updated every 5 days, instead of the pairs  $(x_\tau, y_{\tau+\Delta t})$  ending at  $t - \Delta t$ : an adaptation
- ▣ Strong benchmarks and formal tests instead of one commercial forecast

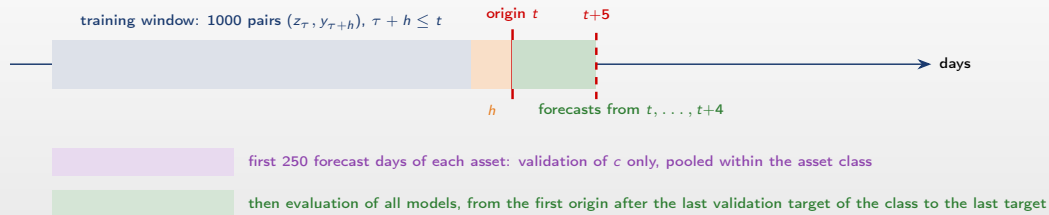
## Hypotheses

- ▣ **H1 (features):** signature terms add information beyond log-HAR
  - ▶ Test: Sig-L vs log-HAR, same target, same equal weights
- ▣ **H2 (weights):** kernel weights improve a given regression
  - ▶ Tests: Sig-LK vs Sig-L and log-HAR-K vs log-HAR
- ▣ **H3 (method):** the full method of Gu et al. beats the HAR family
  - ▶ Tests: Sig-LK vs HAR and vs log-HAR; model confidence set of all seven models
- ▣ **H4 (stress):** any gain is larger in stress periods (COVID-19 crash, 2–30 April 2020, the US tariff shock), where similar past windows should matter most

## Kernel Temperature: One Choice per Class, on a Validation Block

- Scale-free temperature:  $\gamma = c / \text{median}_{\tau}(\delta_{\tau})$  in every window
  - ▶  $c = 1$ : a window at the median distance gets weight  $e^{-1}$  relative to an identical one
- Grid  $c \in \{0.5, 1, 2, 4\}$ , chosen per horizon and asset class on the first 250 forecast days of each asset, pooled within the class
  - ▶ Criterion: mean over the class of  $\text{QLIKE}(c)$  divided by the asset's average over the grid
  - ▶ Evaluation starts after the last validation target of the class, so no tuning target postdates an evaluated origin
- For the 40 stocks the smallest temperature,  $c = 0.5$  (the most nearly equal weights), wins at every horizon for Sig-LK; for futures and FX the choice changes with the horizon: a flat, noisy criterion (all values in the [Appendix](#))

## Rolling Estimation, Step by Step



- Re-estimate every 5 trading days on the last 1000 complete pairs; between re-estimations the coefficients are fixed but the regressors are updated daily
- Kernel distances are recomputed at each re-estimation, against the window ending at the origin  $t$

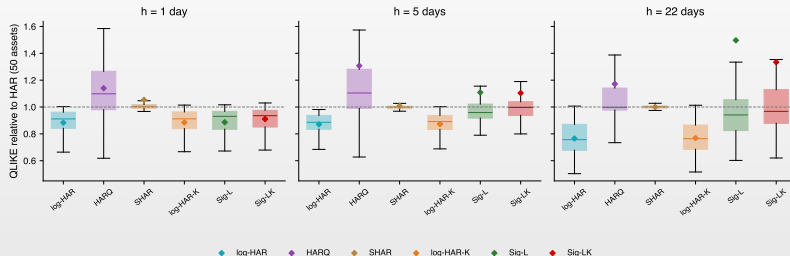
## Evaluation Protocol

- Per asset and horizon: mean QLIKE and MSE; DM (Diebold–Mariano) test with Newey–West variance, bandwidth  $\max(h, \lceil T^{1/3} \rceil)$  [Diebold & Mariano \(1995\)](#)
  - ▶ Rolling estimation, so the test is read as the unconditional GW (Giacomini–White) test; the conditional GW test uses instruments  $(1, d_{t-h})$  [Giacomini & White \(2006\)](#)
- MCS (model confidence set) at 90%,  $T_{\max}$  statistic, stationary bootstrap, 1000 replications [Hansen, Lunde & Nason \(2011\)](#)
  - ▶ Block length  $\max(h, 10)$  days
- Panel: DM test on the cross-sectional mean loss differential (one series per date)
  - ▶ Counting wins over 50 assets: Holm (FWER) and BH (FDR) corrections [Benjamini & Hochberg \(1995\)](#)
- Subperiods: 20 Feb–30 Apr 2020, 2–30 Apr 2025, all other days; the paper's relative error as a descriptive statistic
- 265 929 out-of-sample forecasts per model; stocks are evaluated from January 2020 (the COVID-19 crash is in every stock sample), futures and FX from 2014 (dates in the [Appendix](#))

## A Disclosed Deviation

- Pre-registered: the BPQ insanity filter on all models
  - ▶ A test run of the code on JPM (one asset, before any panel result) showed it replacing correct crash forecasts by the calm in-sample mean
  - ▶ On 12 March 2020 the realised next-day RV of JPM was about 40 times the filtered HAR forecast
- Changed before estimating the models on the 50-asset panel: primary results with positivity only (a forecast  $\leq 0$  becomes the in-sample mean)
  - ▶ The BPQ filter is kept as a robustness check and reported in full
- Rule: a deviation is allowed, a hidden deviation is not; state what was seen when it was changed

## QLIKE Relative to HAR, Asset by Asset



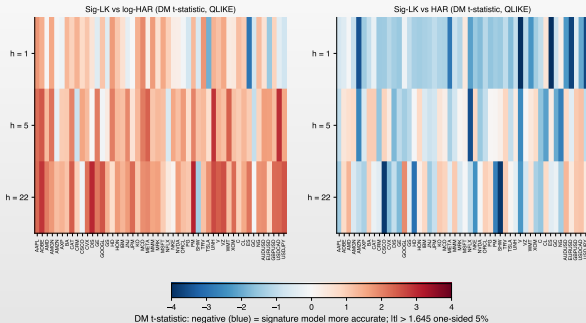
- Boxes: interquartile range over the 50 assets; line: median; diamond: mean; whiskers: 1.5 interquartile ranges
- $h = 1$ : log-HAR, log-HAR-K and Sig-L are close (0.883, 0.885, 0.886); Sig-LK is +4.5% relative to log-HAR: no gain from 36 extra features
- $h = 5, 22$ : Sig-LK falls behind log-HAR (+28.0%, +92.9%); its medians (0.997, 0.968) are much better than its means: a few assets and days drive the loss
- HARQ and SHAR lose to HAR in levels here, because the main design does not apply the BPQ filter (see [Appendix](#): the filter rescues HARQ); first verdict: the method does not beat the simplest log model (a near-tie is not a proof of equal accuracy)

## Asset by Asset: Wins and Losses, Corrected

| Comparison (QLIKE)   | $h = 1$ |       | $h = 5$  |        | $h = 22$ |         |
|----------------------|---------|-------|----------|--------|----------|---------|
| Sig-LK vs log-HAR    | 2/0/0   | 5/0/0 | 0/0/0    | 15/0/2 | 0/0/0    | 27/1/16 |
| Sig-L vs log-HAR     | 3/1/1   | 5/0/0 | 0/0/0    | 9/0/0  | 1/0/0    | 21/0/0  |
| log-HAR-K vs log-HAR | 3/0/0   | 3/0/0 | 3/0/0    | 1/0/0  | 5/0/0    | 6/0/0   |
| Sig-LK vs Sig-L      | 2/0/0   | 5/0/0 | 0/0/0    | 5/0/0  | 2/0/0    | 10/0/0  |
| Sig-LK vs HAR        | 12/5/6  | 0/0/0 | 7/1/1    | 0/0/0  | 8/2/4    | 1/0/0   |
| log-HAR vs HAR       | 21/5/17 | 0/0/0 | 23/13/20 | 0/0/0  | 42/25/42 | 0/0/0   |

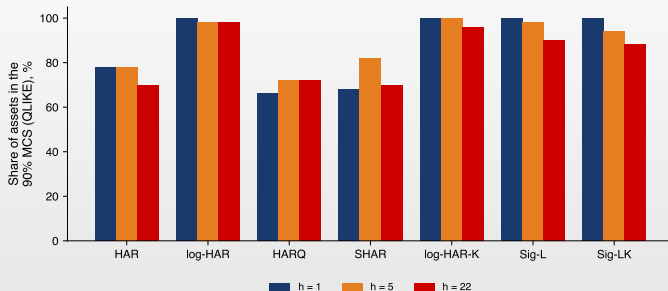
- Cell: significantly better on raw/Holm/BH assets | significantly worse on raw/Holm/BH assets; one-sided DM at 5%, family of 50 assets; Holm controls the FWER under any dependence, BH the FDR under independence or PRDS [Benjamini & Yekutieli \(2001\)](#)
- H1 (signatures): Sig-L beats log-HAR on 3 of 50 assets at  $h = 1$  raw, 1 after Holm; about 2.5 raw wins are expected by chance
- H2 (weights): log-HAR-K vs log-HAR and Sig-LK vs Sig-L: no Holm-significant difference at any horizon
- Against HAR at  $h = 1$ , Sig-LK wins on 12 assets, but log-HAR alone wins on 21: choose the benchmark that isolates the idea

## DM Statistics for Every Asset and Horizon



- Diebold–Mariano  $t$ -statistics of the QLIKE differential; blue: Sig-LK more accurate, red: less accurate
- Against log-HAR (left): pale at  $h = 1$  (blue exceptions ES, TSLA, EURUSD; five assets significantly red before correction), red at  $h = 5, 22$
- Against HAR (right): blue at  $h = 1$  because Sig-LK inherits the log target; one cell such as ES at  $h = 1$  would make signatures look like a clear success

## Model Confidence Sets



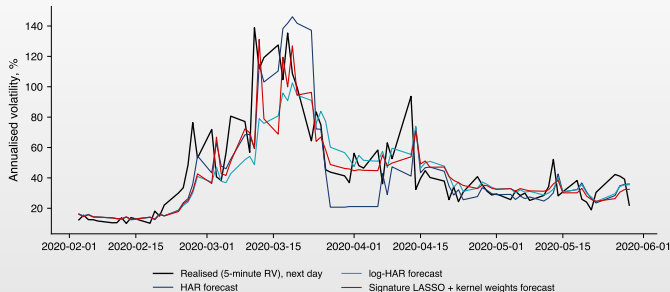
- Share of the 50 assets in which each model belongs to the 90% MCS under QLIKE: log-HAR 100%, 98%, 98%; Sig-LK 100%, 94%, 88% ( $h = 1, 5, 22$ )
- The MCS rarely separates the four log models; HAR, HARQ and SHAR are excluded for a fifth to a third of the assets: the level target is the clearest loser
- Under MSE almost every model stays in the set [Patton \(2011\)](#): small differences in average loss need very long samples [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)

## Panel Tests: Pooling the Evidence

| Comparison: DM $t$ (one-sided $p$ ) | $h = 1$        | $h = 5$        | $h = 22$       |
|-------------------------------------|----------------|----------------|----------------|
| Sig-LK vs log-HAR                   | +1.45 (0.927)  | +1.68 (0.953)  | +1.80 (0.964)  |
| Sig-L vs log-HAR                    | -1.91 (0.028)  | +1.48 (0.931)  | +1.51 (0.935)  |
| log-HAR-K vs log-HAR                | +1.89 (0.971)  | +0.40 (0.654)  | +1.92 (0.973)  |
| Sig-LK vs Sig-L                     | +1.74 (0.959)  | +0.98 (0.837)  | -0.77 (0.221)  |
| log-HAR vs HAR                      | -3.09 (<0.001) | -3.95 (<0.001) | -4.50 (<0.001) |

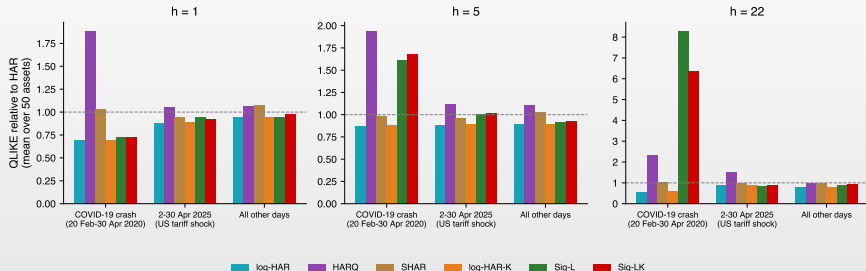
- ▣ Series: the mean QLIKE differential over all assets trading that day; negative  $t$  = first model better
- ▣ Pooled, only the log transform is clearly significant; Sig-L vs log-HAR at  $h = 1$  is borderline ( $p = 0.028$ ), driven by futures
  - For the 40 stocks alone the sign flips:  $t = +1.72$  against Sig-L
- ▣ Conditional GW test, Sig-LK vs log-HAR: 0, 0 and 0 Holm rejections out of 50

## Inside the COVID-19 Crash: JPM, Next Day



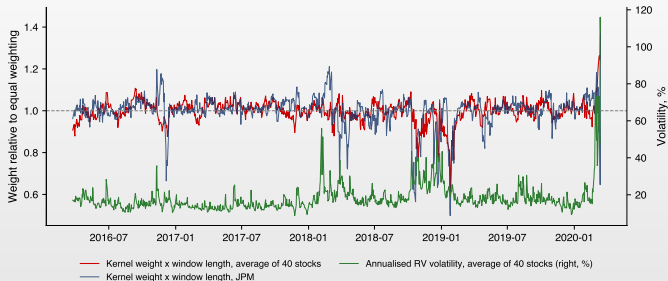
- Annualised volatility  $\sqrt{252F_t}$  of the forecasts and  $\sqrt{252RV_{t+1}}$  of the target, February–May 2020; all models lag the first jump
- Sig-LK reacts slightly faster than log-HAR in mid-March, when the path has already turned; after the peak both decay at similar speeds
- HAR in levels overshoots, then turns negative in late March: the positivity rule replaces it by the in-sample mean (the flat segment)

## Stress Periods vs Normal Times (H4)



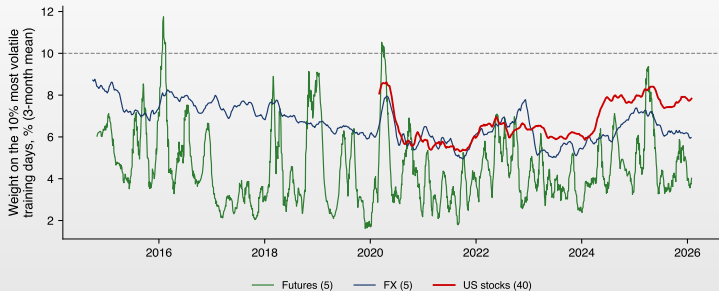
- Mean over assets of QLIKE relative to HAR within each subperiod fixed before the evaluation
- $h = 1$ , COVID-19 crash: log-HAR 0.70, Sig-LK 0.73, HARQ 1.89; 2–30 April 2025 (US tariff shock): 0.88 and 0.92; other days: 0.94 and 0.97
- $h = 22$ , COVID-19 crash: Sig-LK 6.35 and Sig-L 8.30 times the loss of HAR: exactly where similar past windows should help, the signature models fail most; H4 not supported (descriptive, no formal interaction test)

## Kernel Weights at the Origin 16 March 2020



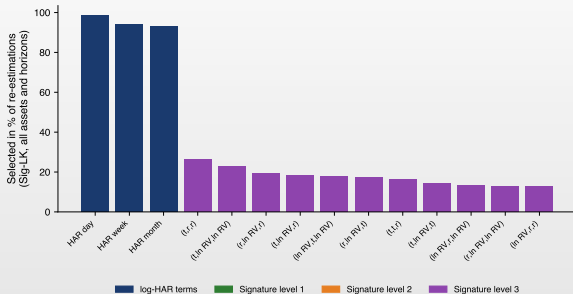
- Weights of Sig-LK ( $h = 1$ ) at the origin 16 March 2020 over the training days (24 Mar 2016 – 13 Mar 2020), times the window length: 1 = equal weighting
- Nearly equal: between 0.61 and 1.43 times  $1/n$ , effective sample size 975 of 1000 (average over the 40 stocks); the highest weights go to the last days before the origin
- The late-2018 sell-off gets *less* weight; weights are almost uncorrelated with log volatility (+0.04): the translation-invariant signature sees shape, not level

## Does the Kernel Ever Favour Stress?



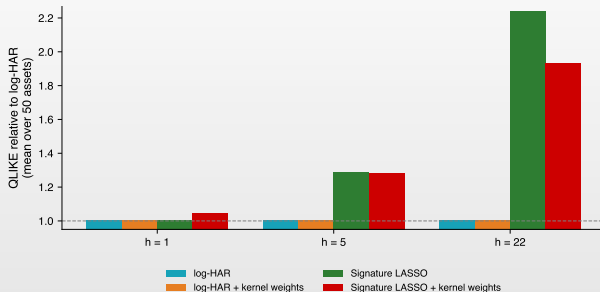
- Weight that Sig-LK ( $h = 1$ ) puts on the 10% most volatile days of its training window, averaged by class, three-month mean
- Stocks: mean 6.7%, between 4.4% and 9.9% (maximum on 17 Mar 2020): always below the 10% of equal weighting; futures and FX show the same pattern
- The mechanism behind the null result for H2: the weights do not implement “learn from past crises”

## What the LASSO Selects



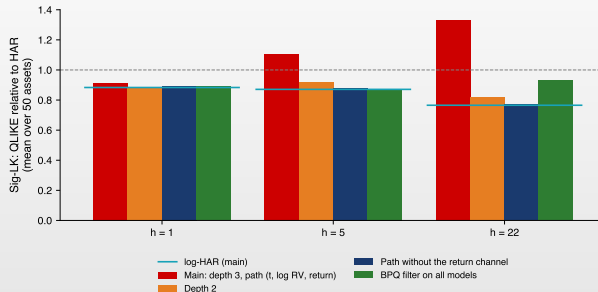
- Share of re-estimations in which a regressor is in the BIC-selected support of Sig-LK;  $t$  = time,  $\ln RV$  =  $\ln RV$ ,  $r$  = cumulative return
- The log-HAR terms are kept in 99%, 94% and 93% of the re-estimations: the LASSO rediscovers HAR; average support 8.7 regressors (Sig-L), 6.7 (Sig-LK)
- The most frequent signature term,  $(t,r,r)$  (27%), and all frequent ones are of level 3: cubic in the increments, far outside the training range in a crash, amplified by  $\exp(\cdot)$

## The $2 \times 2$ Ablation



- Mean over the 50 assets of QLIKE relative to log-HAR: features (signature or not) crossed with weights (kernel or equal)
- Weights alone (log-HAR-K): 1.002, 1.002, 1.004, no effect (average effective sample size 816 of 1000); features alone (Sig-L): 1.004, 1.285, 2.238
- Both (Sig-LK: 1.045, 1.280, 1.929) damp the one-month failure without removing it and worsen  $h = 1$  through crude oil (CL: 2.71); Gu et al. (Table 3) saw a kernel gain in six weekly values of one region

## Robustness of Sig-LK



- Sig-LK relative to HAR (mean over 50 assets) in the main design, in two pre-registered variants and with the BPQ filter (the disclosed deviation); the line marks log-HAR
- Depth 2: 0.922 and 0.820 at  $h = 5, 22$  (main: 1.104, 1.334); path without returns: 0.878 and 0.770: the long-horizon failures come from level-3 terms with the return channel
- The stable variants only match log-HAR (0.871, 0.765): the null result for H1 does not depend on depth or channels, the instability does

## By Asset Class

| Sig-L relative to log-HAR (QLIKE) | $h = 1$ | $h = 5$ | $h = 22$ |
|-----------------------------------|---------|---------|----------|
| Stocks (40)                       | 1.009   | 1.332   | 2.505    |
| Futures (5)                       | 0.965   | 1.084   | 1.206    |
| FX (5)                            | 0.996   | 1.111   | 1.134    |

- The only gain: futures at  $h = 1$  (-3.5%) and, marginally, FX
  - Five futures and one horizon: a candidate for a new, pre-registered study, not a finding of this one
- Stocks, 80% of the panel, show no gain: the pooled result is mostly a stock result

## Limitations and Open Directions

### Limitations of this study

- One design out of many ( $L = 22$ ,  $N = 3$ , three channels, BIC, a four-point grid for  $c$ ): a null result for this design is not one for signatures in general
- Truncated kernel only; no external factors (implied volatility, macro news), central in Gu et al.
- Realized GARCH [Hansen, Huang & Shek \(2012\)](#) and machine-learning benchmarks ([Chapter 13](#)) are outside the comparison; stocks are evaluated over six years only

### Open directions

- A level-aware kernel: basepoint augmentation or the level of  $\ln RV$  in the distance ([Seminar 20](#), C1)
- Signatures of the intraday path (5-minute returns); the PDE kernel [Salvi et al. \(2021\)](#); log-signatures and bounded (tanh) channels against explosive level-3 terms [Morrill et al. \(2020\)](#)
- Cross-asset paths, where Lévy areas measure lead-lag across markets; futures at  $h = 1$  on a longer panel, with this design fixed in advance

## An Open Question

### When does a method transfer across domains?

- Methods travel faster than evidence: a KDD paper on freight rates becomes a candidate for volatility, credit, energy
  - ▶ Which properties of the source problem made the method work, and does the target share them?
- Finance adds a hard filter: 164 teams on the same data disagree more than their standard errors suggest [Menkveld et al. \(2024\)](#)
- Formal version: a transfer succeeds if the gain over the strongest simple benchmark survives a pre-registered set  $\mathcal{S}$  of designs, assets and horizons, with the family corrected
  - ▶ In this chapter:  $\mathcal{S} = 50 \text{ assets} \times 3 \text{ horizons} \times 4 \text{ variants}$ ; Holm/BH were applied per comparison and horizon (families of 50 assets), variants reported descriptively; no joint correction over all 600 cells; the gain did not survive

## The Discovery Loop Applied to This Chapter

- **Read:** an assistant lists every equation, hyperparameter and benchmark of the paper and marks what is not disclosed; each item is checked against the PDF
  - ▶ **Map:** for each component, does it transfer, what must change, which benchmark isolates it
- **Pre-register:** hypotheses, grid, tests and correction fixed in a dated document before the first result; the assistant drafts, the team decides every number
- **Build and test the pieces:** signatures against identities (level 1 = increment, Chen on a split path), not against another library
- **Run the grid, then attack it:** which conclusions survive Holm over 50 assets, which single asset would make a misleading headline?
- **Explain the null:** here flat weights (translation invariance) and explosive level-3 terms (exponential link), both checked with data; the full prompts and the researcher's checklist are in the [▶ Appendix](#)

## Mini-Case: Three Headlines From the Same Runs

- “Signatures beat HAR on 12 of 50 assets” (one-day horizon, raw DM)
  - ▶ True, but log-HAR alone beats HAR on 21: the gain belongs to the log transform
- “Signatures improve E-mini S&P 500 volatility forecasts” (ES,  $h = 1$ , DM  $t = -2.91$  against log-HAR)
  - ▶ One cell out of 150 asset-horizon cells: after Holm, 0 of 50 at  $h = 1$  and none at  $h = 5, 22$
- The headline of the pre-registered evaluation: “no gain over log-HAR; worse at longer horizons; kernel weights inert”
  - ▶ An AI assistant asked to “find where the method works” will produce the first two; the plan protects the third

## Your Turn

- Pick one open direction of this chapter (level-aware kernel, intraday signatures, log-signatures, futures only)
  - ▶ Write  $H_0/H_1$ , the benchmark that isolates it, and the result that would make you drop it
- Use an assistant to list what could go wrong in your design, then write one unit test per item
- Fix the pre-registration in a dated document before the first out-of-sample number; run the grid; report the distribution over  $\mathcal{S}$
- Log every AI error you catch in `AI_ERRORS.md` (Seminar 20, C2 gives typical ones)

## Key Takeaways

- ▣ Signatures turn a window into a feature vector with a theory behind it: Chen's identity, uniqueness, universality
- ▣ Transferred to realised variance, the method of Gu et al. does not beat log-HAR
  - ▶ H1 (signatures add information): not supported, no gain at one day, a loss at a week and a month
  - ▶ H2 (kernel weights help): not supported, weights are almost flat and down-weight past stress; H3: only HAR in levels is beaten, through the log target; H4 (gains in stress): not supported
- ▣ The translation-invariant kernel sees shape, not level; level-3 terms explode in crashes
- ▣ The strongest effect in the study is the oldest one: model the log of RV
- ▣ A pre-registered null result, with ablations and a mechanism, is a finding

## Further Study

### Reading

- ▣ Signatures: Chevyrev & Kormilitzin (2016); Lyons & McLeod (2022); realised measures: Franke, Härdle & Hafner (2019) and Chapter 9

### Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Applied Time Series Analysis with Python
- ▣  Machine Learning in Financial Risk

### Code and data

- ▣ Quantlets: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_20](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_20); data: VOLARE, free for research at [volare.unime.it](https://volare.unime.it)

## Appendix: Features and Scaling

- ▣ Regressors at day  $\tau$  for the signature models: 3 log-HAR terms + 39 signature terms
  - ▶  $\xi_\tau = [\ln \text{RV}_\tau, \ln \text{RV}_\tau^{(w)}, \ln \text{RV}_\tau^{(m)}, \text{Sig}^3(X_{\tau-21:\tau})]$ , path  $X = (\text{time}, \ln \text{RV}, \text{cumulative } r)$ , 22 points
- ▣ Non-time channel  $c$  divided by  $s_c \sqrt{L-1}$ ,  $s_c$  = standard deviation of its daily increments in the training window up to the origin  $t$ 
  - ▶ Time is not rescaled (constant increments,  $s = 0$ ): it runs from 0 to 1; a word  $(i_1..i_k)$  is rescaled by  $\prod_j c_{i_j}$ , so signatures are computed once
- ▣ Before the LASSO every column is standardised with weighted moments of the training window; constant columns (words of time only) are dropped
- ▣ Raw features at  $\tau$  use data up to  $\tau$ ; scales and standardisation use data up to the origin  $t$ : the no-look-ahead cutoff is  $t$

## Appendix: Estimation: Choosing $\lambda$ Without Look-Ahead

- Inside each training window: the whole weighted LASSO path by LARS (least-angle regression)
  - ▶ For every support on the path: weighted OLS refit (step 2 of Gu et al.)
- Pick the support that minimises a BIC-type criterion (Bayesian information criterion) with the Kish effective sample size, a heuristic of this design
  - ▶  $\text{BIC}(S) = n_{\text{eff}} \ln \hat{s}_S^2 + (|S| + 1) \ln n_{\text{eff}}, \quad n_{\text{eff}} = 1 / \sum_{\tau} w_{\tau}^2$
- Why not cross-validation: 50 assets  $\times$  3 horizons  $\times$  hundreds of re-estimations; BIC uses only the window
  - ▶ Kish size corrects for unequal weights, not for serial dependence (overlapping targets,  $h > 1$ ): not a likelihood-based BIC for these data

## Appendix: Kernel Temperatures Chosen on the Validation Block

- ▣ Chosen  $c$  for Sig-LK ( $h = 1, 5, 22$ )
  - ▶ stocks 0.5, 0.5, 0.5; futures 2.0, 1.0, 0.5; FX 0.5, 2.0, 0.5
- ▣ Chosen  $c$  for log-HAR-K ( $h = 1, 5, 22$ )
  - ▶ stocks 0.5, 0.5, 1.0; futures 2.0, 1.0, 4.0; FX 4.0, 2.0, 4.0
- ▣ Grid  $c \in \{0.5, 1, 2, 4\}$ ; the smallest temperature (the most nearly equal weights) wins for the stocks at every horizon; for futures and FX the choice changes with the horizon

## Appendix: What Was Evaluated

| Class   | Evaluation starts | Ends        | Days per asset (median) |
|---------|-------------------|-------------|-------------------------|
| Stocks  | 27 Jan 2020       | 29 Jan 2026 | 1 499                   |
| Futures | 25 Sep 2014       | 29 Jan 2026 | 2 919                   |
| FX      | 27 Aug 2014       | 29 Jan 2026 | 2 952                   |

- 265 929 out-of-sample forecasts per model ( $50 \text{ assets} \times 3 \text{ horizons}$ )
- Stocks start in 2015, so their evaluation begins in January 2020: the COVID-19 crash is in every stock sample
  - Futures and FX start in 2009 and are evaluated from 2014

## Appendix: The Paper's Metric on VOLARE

| Median relative error $ RV - F /RV$ (%) | $h = 1$ | $h = 5$ | $h = 22$ |
|-----------------------------------------|---------|---------|----------|
| HAR                                     | 57.2    | 48.9    | 51.7     |
| log-HAR                                 | 50.4    | 41.9    | 38.7     |
| HARQ                                    | 55.0    | 49.0    | 50.7     |
| SHAR                                    | 57.9    | 49.2    | 51.8     |
| log-HAR-K                               | 49.7    | 41.0    | 38.7     |
| Sig-L                                   | 50.1    | 42.6    | 42.6     |
| Sig-LK                                  | 50.6    | 43.6    | 43.7     |

- Median over assets of the mean relative error: about 40–60%, against 1–2% for freight rates in Gu et al.
- Daily RV is a noisy, heavy-tailed target: the evaluated models have much larger relative errors than those reported for freight rates, without establishing a lower bound on attainable error
  - ▶ Sig-LK forecasts above ten times the in-sample maximum: 8, 29, 36 ( $h = 1, 5, 22$ ), on 10 assets at  $h = 5$ ; log-HAR: none

## Appendix: The Insanity Filter, Revisited

| QLIKE relative to HAR, BPQ filter on all models | $h = 1$ | $h = 5$ | $h = 22$ |
|-------------------------------------------------|---------|---------|----------|
| HAR                                             | 1.000   | 1.000   | 1.000    |
| log-HAR                                         | 0.763   | 0.654   | 0.778    |
| HARQ                                            | 0.995   | 0.912   | 1.014    |
| SHAR                                            | 1.014   | 1.003   | 1.000    |
| log-HAR-K                                       | 0.772   | 0.659   | 0.779    |
| Sig-L                                           | 0.871   | 0.819   | 0.904    |
| Sig-LK                                          | 0.890   | 0.867   | 0.930    |

- The filter rescues HARQ relative to HAR (0.995 at  $h = 1$  instead of 1.140), as BPQ intended
- Sig-LK relative to log-HAR, both filtered: 1.238, 1.424, 1.209 (unfiltered 1.045, 1.280, 1.929): the gap widens at  $h = 1, 5$  and narrows at  $h = 22$
- Filtered forecasts,  $h = 1$ : HAR 180, HARQ 140, Sig-LK 143, log-HAR 29 (of 89 893): the filter hits the signature models far more often than log-HAR

## Appendix: What the Researcher Must Check

- **The method is the paper's:** kernel, weights, refit, as in the cited equations
  - ▶ Test: every design slide cites an equation, table or section of the paper; undisclosed choices are labelled as choices of the study
- **No look-ahead:**  $\lambda$ , scales,  $c$  and features use only data available at the origin
  - ▶ Test: change every value after the origin (for example, multiply future RV by 10); the forecast made at the origin must not change
- **The benchmark isolates the idea**
  - ▶ Test: the  $2 \times 2$  ablation is reported, not only the full model against HAR
- **Losses and tests are the right way round**
  - ▶ Test:  $QLIKE(RV, RV) = 0$  and  $QLIKE(RV, F) > 0$  otherwise; the DM variance is HAC with bandwidth at least  $h$

## Appendix: The Discovery Loop, Prompts (1/2)

- **Read:** extract exactly what the paper did
  - ▶ “From Gu et al. (2024, KDD), list every equation of the estimator, every hyperparameter and its reported value, and the benchmark of each table. Mark what is not disclosed.”
  - ▶ Check each item against the PDF: equation numbers, table numbers, the kernel actually used
- **Map:** what transfers, what must be adapted
  - ▶ “Our target is daily realised variance. For each component of the method, say whether it transfers, what must change, and which benchmark isolates it.”
- **Pre-register:** hypotheses, grid, tests, correction, fixed in a dated document before the first result
  - ▶ The assistant drafts; the team decides every number and signs the dated pre-registration

## Appendix: The Discovery Loop, Prompts (2/2)

- **Build and test the pieces:** signatures against identities, not against another library
  - ▶ “Write unit tests for a depth-3 signature: level 1 equals the increment, the symmetric part of level 2 equals half the outer product, Chen's identity on a split path.”
- **Run the grid, then attack it**
  - ▶ “Here are all 600 runs (50 assets  $\times$  3 horizons  $\times$  4 specifications: main, depth 2, no return channel, BPQ filter). Which conclusions survive Holm over 50 assets? Which single asset or subperiod would make a misleading headline?”
- **Explain the null:** find the mechanism, not an excuse
  - ▶ Here: flat weights (translation invariance) and explosive level-3 terms (exponential link), both checked with data

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